

Technical University of Munich
Chair of AI Planning in Dynamic Environments

Contact-rich Robotics; an optimization perspective
Exercise 1

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Due: 05.05.2026, before the class

1. Download any quadruped robot of your choice from the robot descriptions library, load it into pinocchio, and visualize it following the procedure from tutorial 1. Attach a screenshot of the robot in its neutral configuration as a response to this question.
2. Load the quadruped as a fixed base model and write a simulation loop while ignoring contact forces. Plot the joint trajectories for the following (a) and (b) cases. Make sure to start both from the same initial condition (q, v). You can define the control input $\tau = 0$.

- (a) Use the rigid body dynamics equation to solve for acceleration. Integrate the state forward using Euler integration:

```
v_next = v + a * dt
q_next = q + v * dt
```

- (b) Now repeat the same procedure while using the articulated body algorithm instead of solving for accelerations with the rigid body dynamics equation.

```
a = pin.aba(model, data, q, v, tau)
```

3. Load the quadruped as a floating base model and write a simulation loop while ignoring contact forces. Plot the base and joint trajectories from a).

- (a) Use the articulated body algorithm to solve for acceleration. Integrate the state forward using euler integration while taking into account the floating base configuration manifold:

```
v_next = v + a * dt
q_next = pin.integrate(model, q, v * dt)
```